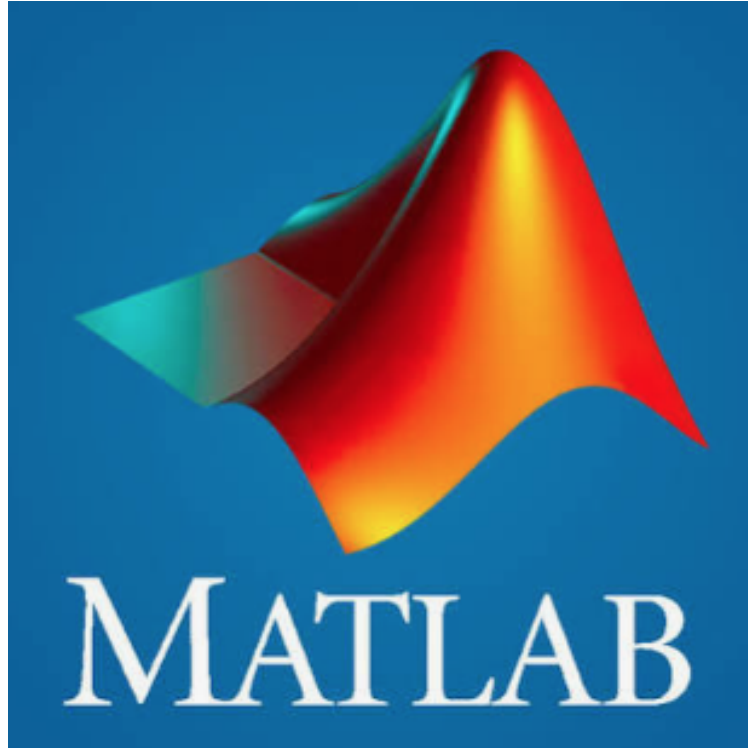
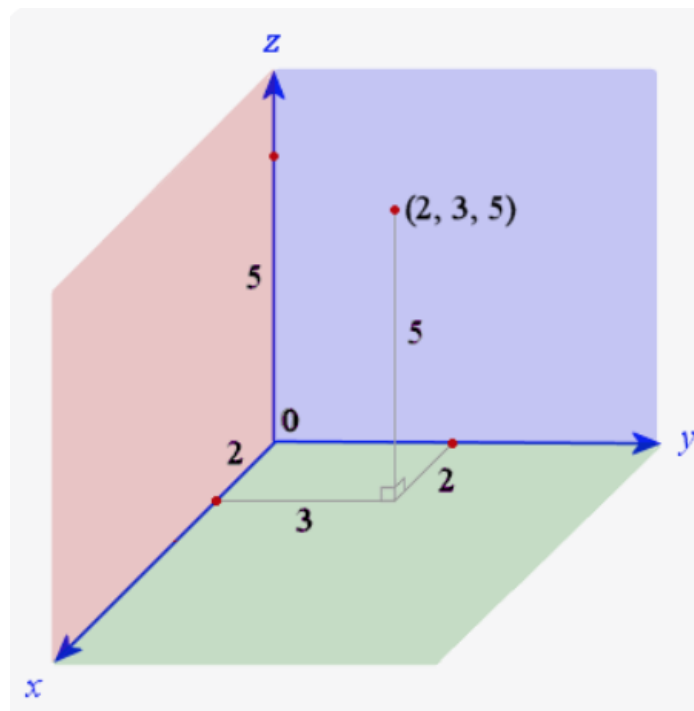


MATH 280 MATLAB Project #1 - Vectors and 3D Space



When the project is completed, rename the end of the file to (Completed Implementation).



PROJECT OUTLINE

Core Objective

Develop a MATLAB project that allows the user to create and visualize points, vectors, lines, planes, cylinders, and quadric surfaces in three-dimensional space while performing common vector operations.

Rather than acting like a calculator, the program acts as a 3D mathematical visualization tool.

Concepts Covered

Chapter 12 - Vectors and the Geometry of Space

- 12.1 3-Dimensional Coordinate Systems
- 12.2 Vectors
- 12.3 Dot Product (vector products)
- 12.4 Cross Product (vector products)
- 12.5 Lines & Planes in Space
- 12.6 Cylinders & Quadric Surfaces

Learning Objectives

By completing this project you will practice:

- MATLAB plotting
- MATLAB graphics
- matrix operations
- vector operations
- user input
- geometric interpretation
- mathematical visualization

Required Features

1. Plotting points in MATLAB

2. Creating vectors in MATLAB
3. Performing vector operations in MATLAB
4. Generating lines in MATLAB
5. Generating planes in MATLAB
6. Generating cylinders in MATLAB
7. Demonstrating the relationship between lines and planes in MATLAB
8. Exploring quadric surfaces in MATLAB
9. Calculating distance in MATLAB
10. Calculating distance in MATLAB

Necessary MATLAB Files For This Project/File Structure

MVC_Project1/

main.m

point3D.m

vectorOperations.m

lineGenerator.m

planeGenerator.m

surfaceGenerator.m

distanceCalculator.m

angleCalculator.m

plotUtilities.m

[README.md](#) (for GitHub)

Useful MATLAB Commands For This Project

plot3()

quiver3()

meshgrid() - used in diff eq

surf()

mesh()

cylinder()
sphere()
hold on - used in diff eq
axis equal - used in diff eq
view()
xlabel() - all axis label commands used in diff eq
ylabel()
zlabel()
legend() - used in diff eq
norm()
dot()
cross()

Additional MATLAB Pseudocode

Vector Magnitude

Receive vector

Compute $\text{sqrt}(x^2 + y^2 + z^2)$

Display magnitude

Dot Product

Receive vectors

dot(u,v)

Determine

acute

obtuse

or perpendicular

Cross Product

Receive vectors

cross(u,v)

Display resulting vector
Plot perpendicular vector

Plane Generation

Receive point
Normal Vector
Generate meshgrid
Solve plane equation
Display surface
Quadric Surface
User selects surface
Generate equations
Compute x,y,z arrays
Plot using surf()
Enable rotation

GUI FUNCTIONALITY

1. Coordinate input boxes
2. Vector dropdown menus
3. Interactive sliders for radius/height
4. Surface selector
5. Rotate View button
6. Reset Plot button
7. Live equation display

MAIN MENU

3D Geometry Visualization Toolkit

1. Plot Points

2. Create Vectors
3. Vector Operations
4. Create Line
5. Create Plane
6. Plot Cylinder
7. Plot Quadric Surface
8. Distance Calculator
9. Exit

multivariable calculus

$\iiint f(x,y,z) \partial f / \partial y \partial f / \partial x 7i + 4j + 3z$